

Technical White Paper: Empirical Verification of a 62-Minute Thermal Soak & Deterministic Zero-Allocation Telemetry Architecture

Client POC evaluation harness (`poc_eval/poc_eval_harness.cpp`) — 153 unattended runs, single host, measured results only.

1. INITIALIZATION

The evaluation target was the client-facing POC harness distributed via `run_poc_eval.bat`, which compiles `poc_eval_harness.cpp` against the single-header `animus.hpp` engine with no third-party benchmarking dependency. Prior to the soak, the build entry point was exercised as a client running it for the first time would encounter it.

That first build attempt failed. `run_poc_eval.bat`'s MSVC compilation step passed `/Fo:"<path>\"` as the object-file output directory — a quoted Windows path ending in a trailing backslash, which `cl.exe`'s argument parser interprets as an escaped quotation mark rather than a path separator. This corrupted the argument boundary and produced `fatal error C1083: Cannot open compiler generated file`. The fix was a one-line change, doubling the trailing backslash inside the quoted path (the standard Windows-argument-parsing workaround), verified by a clean rebuild before the soak began.

Test bench configuration:

PARAMETER	VALUE
CPU	Intel(R) Core(TM) i7-14650HX
Logical cores	24
TSC frequency (calibrated)	2.4192 GHz
Invariant TSC	Yes (CPUID.80000007H:EDX[8])
Core pinning	Enabled, base P-core 4
MPMC producer threads	8
Compiler	MSVC cl.exe, /O2 /std:c++17
Per-run duration	20 s (10 s Phase A / 10 s Phase B)
Inter-run cooldown	3 s

Every thread across both phases is pinned via

`animus::sys::pin_current_thread_to_core_exclusive` — the exclusive variant, which also raises scheduling priority, per this codebase's own documented Phase 14 finding that affinity alone without priority elevation can worsen P99.9 rather than improve it. Baseline calibration (TSC-vs-steady_clock race, 7 samples, median) was performed fresh inside every one of the 153 runs, not reused across runs.

2. SUSTAINED EXECUTION

153 back-to-back runs were executed unattended over 62.1 minutes of wall-clock time. SPSC push-latency P50 held at 21.08 ns for the majority of the run, drifting to a 21.49–21.91 ns band in the final third of the soak window — a small, bounded increase consistent with thermal/scheduler settling under sustained load on a host without dedicated core isolation, not a runaway degradation.

PHASE OF SOAK	SPSC PUSH P50	BEHAVIOR
Runs 1–~100	21.08 ns	Stable
Runs ~100–153	21.49–21.91 ns	Minor upward creep, bounded

MPMC throughput under 8-producer contention held at a median of **7,471,133 pushes/sec** across the full soak, with a measured range of 5,139,660–8,563,594 pushes/sec run-to-run. This variance is consistent with ordinary OS scheduler noise on a non-isolated host and showed no

directional decay over the 62-minute window — the minimum and maximum observations were not clustered at either end of the soak timeline.

METRIC	P50	P90	P99
SPSC ingest latency (median across 153 runs)	21.08 ns	24.39 ns	27.70 ns
Tick-to-telemetry latency (median across 153 runs)	30.59 ns	—	—

Target validation note: every one of the 153 runs reported FAIL against the harness's built-in P50 SPSC ingest target of <15 ns — consistently, from the very first run onward, not as a soak-induced regression. This machine's measured baseline sits several nanoseconds above that target on every single run. The harness reports the measured number as-is; no result was rounded, adjusted, or suppressed to clear the target line.

3. FINAL OUTCOMES & INTEGRITY

Two correctness invariants were tracked on every run via the harness's built-in verification, not inferred after the fact:

INVARIANT	RESULT	STATUS
Heap allocations on hot path (both phases, all runs)	0 bytes	PASS
Event sequence integrity (SPSC, all runs)	153 / 153 intact	PASS

Zero-allocation verification is enforced by a global operator `new/delete` watchdog armed only during the timed measurement region of each phase, converting the "no heap allocation on the hot path" claim into a runtime-checked fact for every one of the 153 runs rather than a static assumption carried over from source inspection. Sequence integrity confirms every pushed event was drained by the consumer exactly once, in order, with no loss or reordering, across all 21.3M+ events pushed per run.

4. OUTLIER TRANSPARENCY

A single SPSC push-latency sample of **~30.03 ms** was recorded during Run #123 (`raw_logs/run_0123.log`). This is reported here without adjustment because it is real, measured data — and because it is precisely the kind of tail event a soak test exists to surface.

RUN POPULATION	OBSERVED MAX LATENCY
Run #123 (1 of 153)	~30,030,989.93 ns (~30.03 ms)
All other 152 runs	30 μs – 1 ms range
P99.9 across the full soak	held steady at ~39–46 ns

The harness pins threads to exclusive cores and elevates their scheduling priority, but this host is not running kernel-level core isolation (`isolcpus`, `nohz_full`, `rcu_nocbs` — Linux-only mechanisms with no Windows equivalent applied here) or NUMA-pinned allocation. Under those conditions, a pinned thread can still be preempted by the OS scheduler for housekeeping work with nowhere else to run until its one core frees up — which is the documented mechanism behind occasional large tail spikes on a non-isolated host. This single event did not recur, cluster, or trend upward across the remaining runs, and every other correctness and latency metric in Run #123 itself was otherwise unremarkable.

5. SUMMARY CARD

Reproduced verbatim from the soak's generated report for reference:

```
+-----+
|          ANIMUS CORE -- CLIENT POC EVALUATION SOAK RESULT          |
+-----+
| Soak duration   : 62.1 min | Runs: 153 | Cooldown: 3s |
| CPU             : Intel(R) Core(TM) i7-14650HX (24 threads) |
| Core pinning    : ENABLED, base P-core 4 |
+-----+
| SPSC INGEST LATENCY (median across 153 runs) |
|   P50   :   21.08 ns |
|   P90   :   24.39 ns |
|   P99   :   27.70 ns |
+-----+
| TICK-TO-TELEMETRY LATENCY |
|   P50   :   30.59 ns |
+-----+
| MPMC THROUGHPUT |
|   Median: 7,471,133 pushes/sec |
|   Range : 5,139,660 - 8,563,594 pushes/sec |
+-----+
| ZERO-ALLOCATION HOT PATH   : 0 bytes -- 153 / 153 runs |
| EVENT SEQUENCE INTEGRITY  : 153 / 153 runs -- 100% intact |
+-----+
```

Animus Technologies (OPC) Private Limited · Low-Latency Systems Group · Full per-run data in [soak_summary.csv](#) · Raw per-run logs in [raw_logs/](#) · All figures measured on this host during this soak window